

# Mean-field models of population activity

Neural mass and field models, Wilson–Cowan

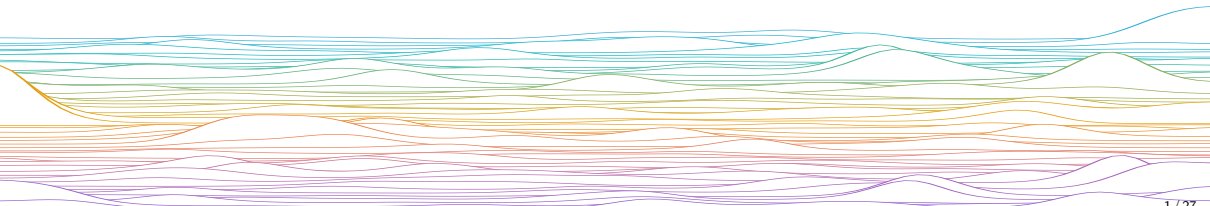
Computational Neuroscience

University of Bristol

mrule

## Learning outcomes:

- Phenomenological mean-population-rate derivation of neural mass models
- Motivate neural mass and neural field models and describe when they are an appropriate modelling choice



## Modelling Scales

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin–Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch–Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,  
Quantitative

Biological  
Realism,  
Data needed  
to identify  
parameters



Computational  
Efficiency,  
Mathematical  
Tractability



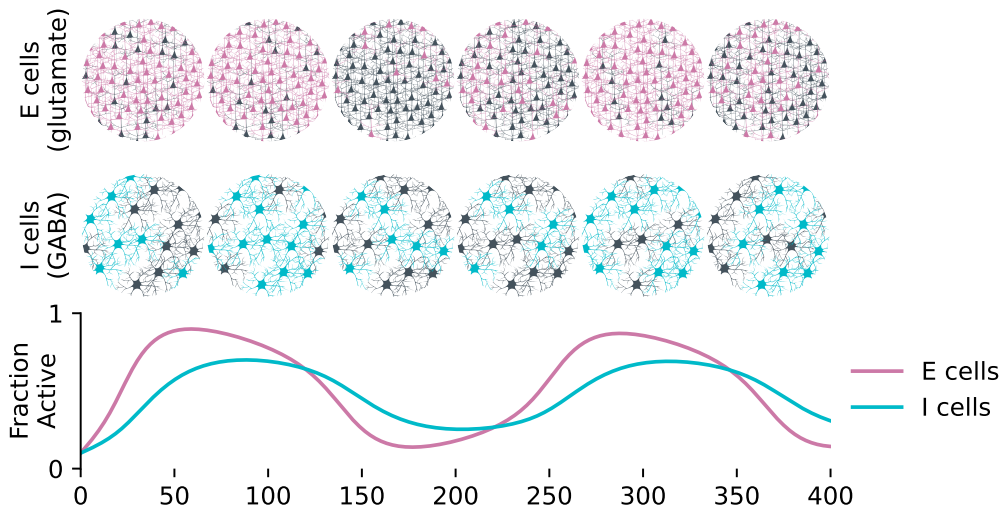
Phenomenological,  
Qualitative

*Neural mass and neural field models*

*This derivation departs slightly from the original*

*Lies from these slides:* The Wilson & Cowan (1972) derivation is more exact: It integrates over the (mean-field) spiking history to track of exactly when in the past neurons spiked (to be more exact about the number that have exited their refractory period). It also includes a 'temporally coarse grained' model that simplifies this integral, but still includes the explicit  $(1 - x)$  terms as in the form on slide 8.

# Mean-Field "Neural Mass" Model



## Neural mass model

Neurons have a maximum firing rate.

- ▶ In time bin of length  $\Delta t < \tau_{\text{refractory}}$ , a neuron either
  - spikes “1” or
  - does not spike “0”

Imagine a neuron as an abstract state  $s(t) \in \{0, 1\}$

- ▶ 0: Quiescent: at rest, able to spike if activated
- ▶ 1: Recently spiked (cannot spike again until entering state 0)
- ▶ Interpret a  $0 \rightarrow 1$  transition as a spike

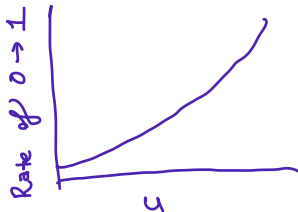
## Neural mass model

In analogy to a mean-field model of an ion channel (open/closed), define state transitions and rates:

- ▶  $1 \rightarrow 0$  with rate  $\beta$
- ▶  $0 \rightarrow 1$  with rate  $\alpha e^u$ , where  $u$  is synaptic activation.

Consider large population of identical neurons  $\{s(t)\}_i$

Let  $x \in [0, 1]$  be the fraction that are 1,  $x(t) = \langle s(t) \rangle$ .



ODE for a mass-action rate model similar to e.g. a gating variable for a channel:

$$\dot{x} = -\beta x + \alpha e^u (1 - x)$$

compare to channel closing  
compare to channel opening

Let's write our ODE in the  $(\tau, x_\infty)$  format

$$\dot{x} = -\beta x + \alpha e^u (1 - x)$$

$$\dot{x} = -[\beta + \alpha e^u]x + \alpha e^u$$

$$\underbrace{\frac{1}{\beta + \alpha e^u}}_{\tau(u)} \dot{x} = -x + \underbrace{\frac{\alpha e^u}{\beta + \alpha e^u}}_{x_\infty(u)}$$

$$\tau(u)\dot{x} = -x + x_\infty \quad \leftarrow$$

$$\tau(u) = \frac{1}{\beta + \alpha e^u}$$

$$x_\infty(u) = \frac{\alpha e^u}{\beta + \alpha e^u}$$



The quantity  $x_\infty(u) = \frac{\alpha e^u}{\beta + \alpha e^u}$  is  $\in [0, 1]$ .

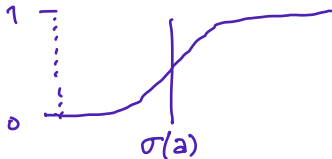
We can re-write  $x_\infty(u)$  using the logistic sigmoid function  $\sigma(q) = [1 + \exp(-q)]^{-1}$

$$\begin{aligned}x_\infty(u) &= \frac{\alpha e^u}{\beta + \alpha e^u} \\&= \frac{1}{1 + \frac{\beta}{\alpha e^u}} \\&= [1 + \frac{\beta}{\alpha} e^{-u}]^{-1} \\&= [1 + e^{-[u - \ln \frac{\beta}{\alpha}]}]^{-1} \\&= \sigma(\underbrace{u - \ln \frac{\beta}{\alpha}}_{\vartheta}) \\&= \sigma(u - \vartheta)\end{aligned}$$

$$\frac{1}{1 + e^{-x}}$$

$\tau(u)\dot{x} = -x + \sigma(u - \vartheta)$





There are many variations of mean-field models with small adjustments for realism or convenience.

E.g. we can approximate  $\tau(u) \sim \tau$  as constant

$$\tau \dot{x} = -x + \sigma(u - \vartheta)$$

We can consider multiple such populations, so that  $x$  is now a vector.

$$\text{diag}[\boldsymbol{\tau}] \dot{\mathbf{x}} = -\mathbf{x} + \sigma(\mathbf{u} - \boldsymbol{\vartheta})$$

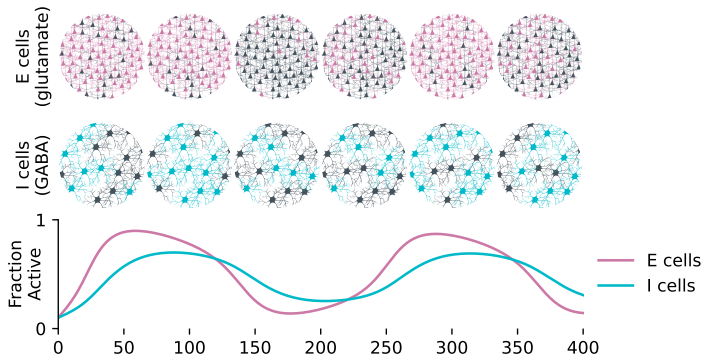
We can add recurrence:  $\mathbf{u} = \mathbf{W}\mathbf{x} + \mathbf{s}$ , where  $\mathbf{s}$  is, for example, some sensory input.

$$\text{diag}[\boldsymbol{\tau}] \dot{\mathbf{x}} = -\mathbf{x} + \sigma(\mathbf{W}\mathbf{x} + \mathbf{s} - \boldsymbol{\vartheta})$$

E.g. two populations, respecting Dale's principle:

$$\tau_E \dot{x}_E = -x_E + \sigma(w_{ee} x_E - w_{ie} x_I - \vartheta_E) \quad \tau_I \dot{x}_I = -x_I + \sigma(w_{ei} x_E - w_{ii} x_I - \vartheta_I)$$

### Mean-Field "Neural Mass" Model



≈ 80% of neurons in cortex are excitatory "E", 20% inhibitory "I";

## Wilson–Cowan model (1972)

$$\tau_E \dot{x}_E = -x_E + \sigma(w_{ee} x_E - w_{ie} x_I - \vartheta_E)$$

$$\tau_I \dot{x}_I = -x_I + \sigma(w_{ei} x_E - w_{ii} x_I - \vartheta_I)$$



Hugh Wilson



Jack Cowan

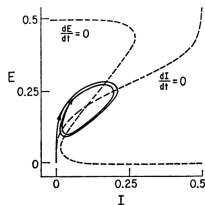


FIGURE 11 a

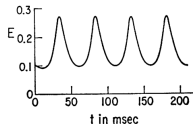


FIGURE 11 b

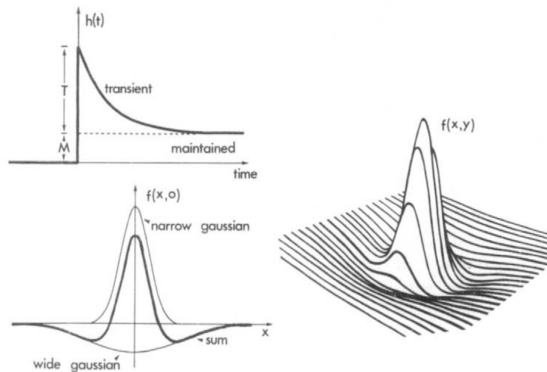
FIGURE 11 a Phase plane showing limit cycle trajectory in response to constant stimulation  $P = 1.25$ . Dashed lines are isoclines. Parameters:  $c_1 = 16, c_2 = 12, c_3 = 15, c_4 = 3, a_e = 1.3, \theta_e = 4, a_i = 2, \theta_i = 3.7, r_e = 1, r_i = 1$ .

FIGURE 11 b  $E(t)$  for limit cycle shown in Fig. 11 a.  $\tau = 8$  msec.

**Lies from these slides:** The Wilson & Cowan (1972) derivation is more exact: It integrates over the (mean-field) spiking history to track of exactly when in the past neurons spiked (to be more exact about the number that have exited their refractory period). It also includes a ‘temporally coarse grained’ model that simplifies this integral, but still includes the explicit  $(1 - x)$  terms as in the form on slide 8. Also I’m using different names for the parameters to keep things consistent with the slides. I also cheated a bit to make the emergence of the logistic sigmoid nonlinearity  $\sigma(\cdot)$  appear more natural than it really is—in practice the activation function (AKA firing-rate nonlinearity) is somewhat arbitrary.

*Neural field models*

## Recall: Modelling rate coding in neural populations

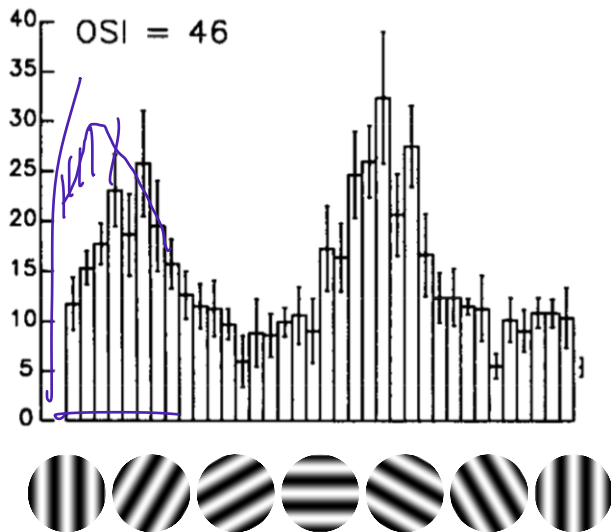


Rodieck, *Vision Res*, 1965

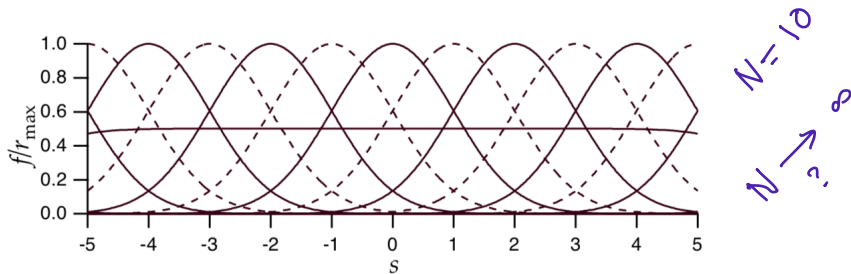
$$f(x, y) = \frac{g_1}{\pi\sigma_1^2} \exp\left(-\frac{x^2 + y^2}{\sigma_1^2}\right) - \frac{g_2}{\pi\sigma_2^2} \exp\left(-\frac{x^2 + y^2}{\sigma_2^2}\right)$$

**Difference of Gaussians** captures the 'ON' centre and 'OFF' surround *typical of retinal ganglion cells*.

## Recall: Tuning Curves



## Recall: Modelling rate coding in neural populations



Dayan & Abbott, 2001

A stimulus ' $s$ ' can be represented by the collective activity of a population of neurons, each with a different preferred stimulus (i.e., their tuning curves are centred at different locations).

In some cases (e.g. when a stimulus space varies smoothly and continuously is well-covered by a neural population), it can make sense to pretend that we have an infinite population of neurons with a cell for each possible  $S$ . This is a *neural field model* and we will cover this in more detail later.



## *Wilson–Cowan–Amari Neural Field Models*

Wilson & Cowan (1973)

- ▶ “A mathematical theory of the functional dynamics of cortical and thalamic nervous tissue”
- ▶ based on spatially extended Partial Differential Equations (PDEs)

Shun'ichi Amari (1977)

- ▶ “Dynamics of pattern formation in lateral-inhibition type neural fields”
- ▶ based on spatially extended Integro Differential Equations (IDEs)



*Hugh Wilson*



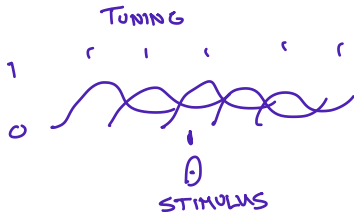
*Jack Cowan*



*Shun'ichi Amari*

On this slide, the lecturer should draw a sheet of neurons, which have a range of tuning to various orientations similar to the Chapman & Stryker (1993) figure on earlier slides. The lecturer should explain the idea that if the population coverage of the tuning space is dense enough, we might imagine that it is possible to define a neural mass population model for each preferred orientation  $\theta_0$ . Granted, interpreting this mean-field model literally requires placing infinitely many neurons in an infinitesimally wide tuning bin  $d\theta$ . This is slightly absurd. It is possible to derive a version that accounts for the finite population size, but this added realism is not necessary.

$N=5$



RESPONSE



$N=\infty$



# CENTRE - SURROUND LATERAL INHIBITION

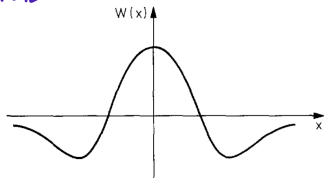


Fig. 2. Weighting function  $w(x)$  of a lateral-inhibition type neural field

Amari 1977 Figure 2

## Example neural field



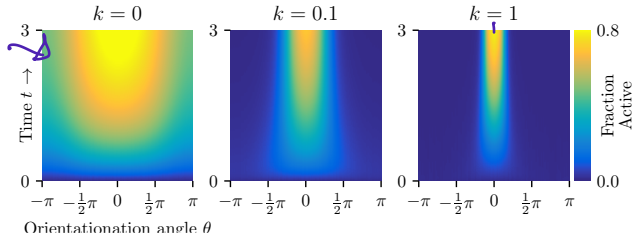
Let's say ...

- The encoded variable is orientation  $\theta$
- Input drive for  $\theta^*$ :  $s(\theta) = \cos(\theta - \theta^*) + 1$
- Lateral interactions:  $w(\theta, \theta') = \cos(\theta - \theta') - 0.95$

$$\dot{x}(\theta) = -x(\theta) + \sigma \left[ k \cdot \int_0^{2\pi} w(\theta, \theta') x(\theta') d\theta' + s(\theta) \right]$$

LESS PRECISE INPUT?

Turning on local, recurrent lateral inhibition:



REINFORCE  
Similar stims  
SUPPRESS DIFFERENT  
SHARPENING VIA  
SURROUND SUPPRESSION

[github.com/michaelerule/neuralfield](https://github.com/michaelerule/neuralfield)

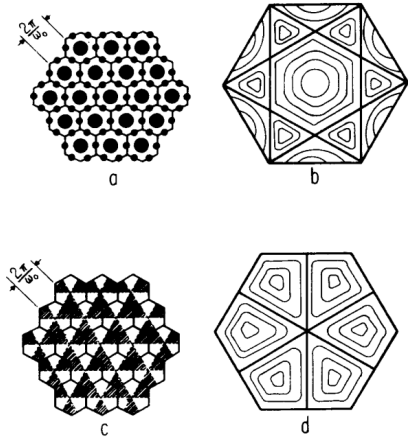
Ermentrout & Cowan (1979)  
 “A Mathematical Theory of Visual Hallucination Patterns”



Bard Ermentrout



Jack Cowan



**Fig. 10a–d.** Hexagonal lattice patterns [shaded areas positive and contours represent  $E(x, y) = \text{const}$ ]. Hexagonal pattern,  $\theta_1 = \theta_2 = \theta_3 = 0$ . **b** Contours if (a) for single hexagon. **c** Hexagonal pattern,  $\theta_1 = 90^\circ, \theta_2 = \theta_3 = 0$ . **d** Contours of (c) for single hexagon

*But, before going further here, we should review the visual system (future slides)*

### When can I use a neural mass/field model?

- ▶ A population-rate code should be a sensible approximation of the activity & computation
- ▶ If helps if assumptions about a “mean rate” existing are true:
  - The neural population should be large
  - Connections within this population should be reasonably dense and random
  - Neurons should be “similar”, or show heterogeneity that is statistically manageable
  - If using the coarse-grained equations here: Excess synchrony might be an issue (but, there are other more detailed mean-field models that can handle this).

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### Useful modelling scale for many collective phenomena:

- ▶ Things that don't depend on specific neuronal coding roles:
  - Spatiotemporal waves, patterns
  - Population oscillations
  - Epilepsy
- ▶ Systems that contain large populations of similarly-tuned neurons
  - Early visual system (retina, V1)
  - Encoding head-direction / position
  - Sometimes similarly-tuned populations are spatially nearby (topography), but this is not mandatory



## Summary

A \_\_\_\_\_ model lumps together activity of a neural population to write a simpler low-order ODE for population-averaged quantities.

The \_\_\_\_\_ coarse-grained model defines a two-dimensional ODE for the fractions of active excitatory (E) and inhibitory (I) cells.

A \_\_\_\_\_ model takes a continuum limit of a firing-rate model to describe neural population responses to, e.g. edge orientation  $\theta$  as a partial and/or integro differential equation.

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A neural mass (or mean-field) model lumps together activity of a neural population to write a simpler low-order ODE for population-averaged quantities.

The Wilson–Cowan coarse-grained model defines a two-dimensional ODE for the fractions of active excitatory (E) and inhibitory (I) cells.

A neural field model takes a continuum limit of a firing-rate model to describe neural population responses to, e.g. edge orientation  $\theta$  as a partial and/or integro differential equation.

*end*



